

Regret Minimization in Average-Reward Markov Decision Processes

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1 Introduction

In this note, we study online reinforcement learning in average-reward MDPs. Specifically, we are interested in a setting where the learning performance is assessed through the notion of regret. The corresponding RL problem is sometimes referred to as **regret minimization**.

Notations. We introduce notations that will be used throughout. Given a set A , $\Delta(A)$ denotes the simplex of probability distributions over A . $\mathbb{I}\{A\}$ denotes the indicator function of event A , namely it equals 1 if A occurs, else 0. $\mathbb{N}$ and $\mathbb{R}$ denotes the set of naturals and reals, respectively. Unless specified, $\|\cdot\|$ is used to denote a generic norm. We write $X \sim p$ to indicate that X is drawn randomly from distribution p . Finally, given a set A , $|A|$ denotes the cardinality of A , and a function $f : A \rightarrow \mathbb{R}^{|A|}$ could interchangeably be viewed as a $|A|$ -dimensional vector.

2 Problem Formulation

Let us consider a finite infinite-horizon average-reward MDP $M = (\mathcal{S}, \mathcal{A}, P, R)$, and let S and A be the respective cardinality of $\mathcal{S}$ and $\mathcal{A}$. In a previous note, we discussed that a gain-optimal policy can be characterized by the optimal Bellman equation, which can be solved by, e.g., value iteration.

We consider the problem of reinforcement learning in M , where the MDP is initially unknown to the agent. The agent's goal is to maximize the T -step total reward $\sum_{t=1}^T r_t$ starting from an initial state decided by nature, potentially for any value of T that could be unspecified. The agent knows only the state-action of M – i.e., she knows what actions are available at any state – and may know some mild information on the reward function (e.g., the range of reward values). Hence, to find strategies maximizing the said objective, the agent must rely on the collected experience which is collected in a growing sequence $h_t = (s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_t)$. Specifically, in the online setting, it is assumed that there is no separate training phase, in the sense that *the agent starts learning from the scratch, and has a single life, namely all the mistakes she makes count in her performance*.

Relating the objective to optimal gain (and friends). We would like to relate the objective function, $\sum_{t=1}^T r_t$, to quantities such as gain, etc. This may not be an easy task if the policy changes over time. But for now let's assume that the policy is fixed. Under a stationary policy π , $\sum_{t=1}^T r_t$ approaches $Tg^\pi(s_1)$, as T grows large, where s_1 denotes the starting state. Let's consider a gain-optimal policy π^* , under which the sequence of rewards is denoted by $(r_t^*)_{t \geq 1}$. Then, $\frac{1}{T} \sum_{t=1}^T r_t^* \rightarrow_{T \rightarrow \infty} g^*$. Further, the following

non-asymptotic bounds hold:

$$(i) \quad \mathbb{E} \left[\sum_{t=1}^T r_t^* \middle| s_1 = s \right] = Tg^* + \mathcal{O}(\text{sp}(b^*)),$$

$$(ii) \quad \sum_{t=1}^T r_t^* \leq Tg^* + \mathcal{O}(\text{sp}(b^*)\sqrt{T \log(T/\delta)}), \quad \text{w.p.} \geq 1 - \delta.$$

For derivation, refer to [1]. Therefore, not only $\sum_{t=1}^T r_t^*$ mimics Tg^* asymptotically, but also the mismatch between the two can be controlled *non-asymptotically*, and even with high probability. This result is also valid under any stationary policy π that induces a recurrent Markov chain in the MDP M .

This suggests that the agent can resort to learning a gain-optimal policy $\pi^* \in \Pi^{\text{SD}}$ in M :

$$\pi^*(s) \in \arg \max_{\pi^* \in \Pi^{\text{SD}}} g^\pi(s)$$

Hence, in online average-reward RL, the goal can be set to learn π^* from collected experience.

Learning performance. That the agent in online RL must learn an optimal behavior from scratch, and only from the collected experience, reveals the need for a tradeoff between **exploration** and **exploitation**: At each step she faces the dilemma as to whether she must act consistently with her knowledge, captured via h_t , to maximize the objective – i.e., to exploit –, or whether she shall try an action that is never tried, or sufficiently enough, to better learn its consequence –i.e., to explore. In fact, the key challenge in online RL is to implement the right balance between exploration and exploitation.

Many offline algorithms could be somewhat tailored to work in the online setting with some tricks. However, they may fail to yield a delicate exploration-exploitation tradeoff in a robust and guaranteed way. Therefore, in online RL, we need performance metrics to measure the quality of exploration-exploitation tradeoff. The performance of an online RL algorithm $\mathbb{A}$ can be measured through the following metrics:

- **Convergence:** Whether $\mathbb{A}$ converges to an optimal (or near-optimal) policy.
- **PAC Sample Complexity:** The number of steps where the value of the current policy output by $\mathbb{A}$ is not near-optimal with high-probability.
- **Regret:** The amount of reward lost due to choosing sub-optimal actions by $\mathbb{A}$.

In online average-reward RL, the **regret** of an algorithm $\mathbb{A}$ is the difference in cumulative reward between an optimal policy π^* and the learning $\mathbb{A}$:

$$\mathfrak{R}(\mathbb{A}, T) := \sum_{t=1}^T r_t^* - \sum_{t=1}^T r_t^{\mathbb{A}}$$

where $(r_t^*)_{t \geq 1}$ and $(r_t^{\mathbb{A}})_{t \geq 1}$ denote, respectively, the sequence of reward under an optimal policy π^* and the learning algorithm in question $\mathbb{A}$. More precisely,

$$\forall t: \quad a_t^{\mathbb{A}} = \mathbb{A}(h_t), \quad r_t^{\mathbb{A}} \sim R(s_t^{\mathbb{A}}, a_t^{\mathbb{A}}), \quad s_{t+1}^{\mathbb{A}} \sim P(\cdot | s_t^{\mathbb{A}}, a_t^{\mathbb{A}})$$

Further,

$$\forall t: \quad r_t^* \sim R(s_t^*, \pi^*(s_t^*)), \quad s_{t+1}^* \sim P(\cdot | s_t^*, \pi^*(s_t^*))$$

Note that $\mathfrak{R}(\mathbb{A}, T)$ is a random variable and we wish to control its expectation or its deviation. An algorithm $\mathbb{A}$ is a **no-regret learning algorithm** if either $\mathbb{E}[\mathfrak{R}(\mathbb{A}, T)] = o(T)$ or $\mathfrak{R}(\mathbb{A}, T) = o(T)$ with high probability.

Regret is directly connected to the agent's goal (maximizing $\sum_{t=1}^T r_t$). Alternatively, the objective of the agent is to minimize the regret. In the case of online average-reward RL (and undiscounted RL, in

general), regret is more meaningful than other performance measures, as it is directly connected to the agent's objective $\sum_{t=1}^T r_t$.

We can show that, with probability greater than $1 - \delta$,

$$\begin{aligned}\mathfrak{R}(\mathbb{A}, T) &:= \sum_{t=1}^T r_t^* - \sum_{t=1}^T r_t^{\mathbb{A}} \\ &\leq Tg^* - \sum_{t=1}^T r_t^{\mathbb{A}} + \mathcal{O}\left(\text{sp}(b^*)\sqrt{T\log(T/\delta)}\right)\end{aligned}$$

Hence, controlling $Tg^* - \sum_{t=1}^T r_t^{\mathbb{A}}$ leads to controlling $\mathfrak{R}(\mathbb{A}, T)$. Therefore, for ease of analysis and evaluation, we restrict attention to the following simplified notion of regret (while overloading the notation):

$$\mathfrak{R}(\mathbb{A}, T) = Tg^* - \sum_{t=1}^T r_t^{\mathbb{A}}$$

This notion is also consistent with the gain-optimal policy being sought by the agent.

In what follows, when it is clear from the context that the learning algorithm is $\mathbb{A}$, we drop the dependence of various quantities on $\mathbb{A}$ to avoid clutter.

Definition 1 An algorithm $\mathbb{A}$ is called **no-regret** if there exists a deterministic function f with $\frac{f(T)}{T} \rightarrow_{T \rightarrow \infty} 0$, such that one of the following holds:

$$\begin{aligned}\mathbb{E}[\mathfrak{R}(\mathbb{A}, T)] &\leq f(T) \\ \mathfrak{R}(\mathbb{A}, T) &\leq f(T) \quad \text{with high probability.}\end{aligned}$$

f could be MDP-dependent but must be deterministic. Note that a high-probability bound on $\mathfrak{R}(\mathbb{A}, T)$ implies a bound on $\mathbb{E}[\mathfrak{R}(\mathbb{A}, T)]$, but not the other way around.

3 The UCRL2 Algorithm

In this section, we briefly present the UCRL2 algorithm, a well-known algorithm introduced by Jaksch et al. [2] for online average-reward RL in finite MDPs, admitting regret guarantees with high-probability.

Specifically speaking, UCRL2 is a model-based RL algorithm for regret minimization in average-reward *communicating* MDPs. UCRL2 maintains a delicate exploration-exploitation tradeoff by implementing a form of *the optimism in the face of uncertainty* principle (or for short, the optimism principle). The term UCRL2 is short for *Upper Confidence Reinforcement Learning*, which indicates that the optimism principle is implemented by maintaining upper confidences on average-reward of policies.

By a slight abuse of notation, let us assume that the reward function R is supported on $[0, 1]$,¹ and by a slight abuse of notation, let us use R to also denote the expected value of the reward function R .

In order to implement the optimism principle, UCRL2 maintains confidence sets for P and R as follows. Specifically, at each time $t \geq 1$, the algorithm maintains $\hat{R}_t$ and $\hat{P}_t$, the empirical mean reward and transition functions built using the observations available up to the of time slot t . More precisely, let $N_t(s, a)$ denote the number of times action a was sampled in state s up to the beginning of time slot t (i.e., excluding the ones collected at t):

$$N_t(s, a) = \sum_{t'=1}^{t-1} \mathbb{I}\{s_{t'} = s, a_{t'} = a\}.$$

¹This assumption can be relaxed to include many other interesting rewards (e.g., those following a Gaussian distribution) as long as some mild regularity assumptions are met.

Then, the empirical estimates $\hat{R}_t$ and $\hat{P}_t$ are defined as follows: For all (s, a) ,

$$\begin{aligned}\hat{R}_t(s, a) &= \frac{1}{\max\{N_t(s, a), 1\}} \sum_{t'=1}^{t-1} r_{t'} \mathbb{I}\{s_{t'} = s, a_{t'} = a\}, \\ \hat{P}_t(s'|s, a) &= \frac{1}{\max\{N_t(s, a), 1\}} \sum_{t'=1}^{t-1} \mathbb{I}\{s_{t'} = s, a_{t'} = a, s_{t'+1} = s'\}, \quad \forall s'.\end{aligned}$$

For a given confidence parameter $\delta \in (0, 1)$ (which will be an input parameter chosen by the agent), UCRL2 defines the following confidence sets: For each $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$\mathbf{C}_{t,\delta}^{\text{UCRL2}}(s, a) = \left\{ P' \in \Delta(\mathcal{S}) : \|\hat{P}_t(\cdot|s, a) - P'\|_1 \leq \sqrt{\frac{14S}{\max\{N_t(s, a), 1\}} \log\left(\frac{2At}{\delta}\right)} \right\}, \quad (1)$$

$$\mathbf{C}_{t,\delta}'^{\text{UCRL2}}(s, a) = \left\{ R' \in [0, 1] : |\hat{R}_t(s, a) - R'| \leq \sqrt{\frac{3.5}{\max\{N_t(s, a), 1\}} \log\left(\frac{2SA t}{\delta}\right)} \right\}. \quad (2)$$

We further define

$$\mathbf{C}_{t,\delta}^{\text{UCRL2}} = \cap_{s,a} \mathbf{C}_{t,\delta}^{\text{UCRL2}}(s, a), \quad \mathbf{C}_{t,\delta}'^{\text{UCRL2}} = \cap_{s,a} \mathbf{C}_{t,\delta}'^{\text{UCRL2}}(s, a). \quad (3)$$

The confidence set $\mathbf{C}_{t,\delta}^{\text{UCRL2}}(s, a)$ is constructed using Weissman's concentration inequality [3] for categorical distributions (see Appendix A), which is combined with a union bound to account for randomness in the number $N_t(s, a)$ of observations. The confidence set $\mathbf{C}_{t,\delta}'^{\text{UCRL2}}(s, a)$ is constructed using Hoeffding's inequality for bounded distributions (in $[0, 1]$) combined with a union bound to account for randomness in the number $N_t(s, a)$ of observations. The confidence sets are defined to guarantee that they trap the true P and R with high probability for all $t \geq 1$.

At a high level, at time $t \geq 1$ and for an input confidence parameter δ , UCRL2 maintains the set of MDPs

$$\mathcal{M}_{t,\delta} = \left\{ M' = (\mathcal{S}, \mathcal{A}, R', P') : P' \in \mathbf{C}_{t,\delta}^{\text{UCRL2}}, R' \in \mathbf{C}_{t,\delta}'^{\text{UCRL2}} \right\},$$

It then implements the optimism principle by trying to solve

$$\max_{M' \in \mathcal{M}_{t,\delta}} \max_{\pi \in \Pi_{M'}^{\text{SD}}} g_{M'}^{\pi},$$

where $g_{M'}^{\pi}$ denotes the average-reward (or gain) of policy π in MDP M' . This is carried out approximately by Extended Value Iteration (EVI) that finds a $\frac{1}{\sqrt{t}}$ -optimal solution to the above optimization problem in an efficient manner —EVI is presented later. In other words, EVI finds a policy π_t and an MDP $\widetilde{M}_t$ such that

$$g_{\widetilde{M}_t}^{\pi_t} \geq \max_{M' \in \mathcal{M}_{t,\delta}} \max_{\pi \in \Pi_{M'}^{\text{SD}}} g_{M'}^{\pi} - \frac{1}{\sqrt{t}}.$$

Finally, UCRL2 does not recompute π_t at each time step. Instead, it proceeds in *internal episodes*, indexed by $k \in \mathbb{N}$, where the above policy computation only takes place at the starting time of each episode. Concretely, letting t_k denote the starting time of episode k , the algorithm computes $\pi_k := \pi_{t_k}$ and applies it until $t = t_{k+1} - 1$, where t_{k+1} is the first time step where the number N_t of observations gathered on some state-action pair is doubled since the time the episode k is started (i.e., when compared to N_{t_k}). In other words, episode k starts at $t = t_k$ and continues until there is some (s, a) for which we collect $N_{t_k}(s, a)$ many observations *only during* k . Mathematically, the sequence $(t_k)_{k \geq 1}$ is defined as follows: $t_1 = 1$, and for all $k > 1$,

$$t_k = \min \left\{ t > t_{k-1} : \max_{s,a} \frac{1}{N_{t_{k-1}}(s, a)} \sum_{t'=t_{k-1}}^{t-1} \mathbb{I}\{(s_{t'}, a_{t'}) = (s, a)\} \geq 1 \right\},$$

Algorithm 1 UCRL2 [2]

input: State-space $\mathcal{S}$, action-space $\mathcal{A}$, and confidence level $\delta \in (0, 1)$
initialization: For all (s, a) , set $N_0(s, a) = 0$ and $v_0(s, a) = 0$. Set $t_0 = 0$, $t = 1$, $k = 1$.
// $v_k(s, a)$ counts the visits to (s, a) inside episode k
for episodes $k = 1, 2, \dots$ **do**
 Set $t_k = t$
 Set $N_{t_k}(s, a) = N_{t_{k-1}}(s, a) + v_{k-1}(s, a)$ for all (s, a)
 Compute empirical estimates $\hat{R}_{t_k}(s, a)$ and $\hat{P}_{t_k}(\cdot|s, a)$ for all (s, a)
 Compute $\pi_k = \text{EVI}(\mathcal{M}_{t, \delta}, \frac{1}{\sqrt{t_k}})$ *//See Algorithm 2*
 // π_k is a $\frac{1}{\sqrt{t_k}}$ -optimal policy in $\mathcal{M}_{t, \delta}$.
 Set $v_k(s, a) = 0$ for all (s, a)
 //Apply π_k until $N_{t_{k-1}}(s, a)$ for some (s, a) is doubled.
 while $v_k(s_t, \pi_k(s_t)) < \max\{1, N_{t_k}(s_t, \pi_k(s_t))\}$ **do**
 Observe the current state s_t , play action $a_t = \pi_k(s_t)$, and receive reward $r(s_t, a_t)$
 Set $v_k(s_t, a_t) = v_k(s_t, a_t) + 1$
 Set $t = t + 1$
 end while
end for

EVI: Value Iteration for Bounded-Parameter MDPs Algorithm 2 presents the pseudo-code of the Extended Value Iteration algorithm (EVI). Generically, EVI takes as input an accuracy ε and a *bounded-parameter MDP* $\mathcal{M}$,² and returns an ε -optimal policy among all MDPs in $\mathcal{M}$. Here, we present EVI generically for when $\mathcal{M}$ taking the following form:

$$\mathcal{M} = \left\{ M' = (\mathcal{S}, \mathcal{A}, P', R') : P'(\cdot|s, a) \in \mathcal{C}(s, a) \text{ and } R'(s, a) \in \mathcal{C}'(s, a), \forall s, a \right\}.$$

For example, for UCRL2, $\mathcal{C}(s, a)$ and $\mathcal{C}'(s, a)$ are defined according to (1) and (2), and $\varepsilon = \frac{1}{\sqrt{t_k}}$.

Algorithm 2 EVI($\mathcal{M}, \varepsilon$) [2]

initialization: Select $V_0 \in \mathbb{R}^{\mathcal{S}}$ arbitrarily. Set $n = -1$.
repeat:
 Increment n .
 Compute, for each (s, a) ,

$$R'(s, a) = \max \left\{ \lambda \in \mathcal{C}'_{t, \delta}(s, a) \right\}$$

$$P'(\cdot|s, a) \in \operatorname{argmax} \left\{ \sum_{x \in \mathcal{S}} q(x) V_n(x) : q \in \mathcal{C}_{t, \delta}(s, a) \right\}$$

 //See Algorithm 3 for computing P' .
 Update, for each $s \in \mathcal{S}$,

$$V_{n+1}(s) = \max_{a \in \mathcal{A}_s} \left(R'(s, a) + \sum_{x \in \mathcal{S}} P'(x|s, a) V_n(x) \right)$$

until $\operatorname{sp}(V_{n+1}(s) - V_n(s)) < \varepsilon$
output:

$$\pi^{\text{EVI}}(s) = \operatorname{argmax}_a \left(R'(s, a) + \sum_{x \in \mathcal{S}} P'(x|s, a) V_n(x) \right), \quad s \in \mathcal{S}$$

We now discuss solving the maximization problems in EVI. Suppose the sets $\mathcal{C}(s, a)$ and $\mathcal{C}'(s, a)$

²The set $\mathcal{M}_{t, \delta}$ maintained by UCRL2 is a special case of a bounded-parameter MDP.

take the following generic forms:

$$\mathcal{C}(s, a) = \left\{ q \in \Delta(\mathcal{S}) : \|\widehat{P}(\cdot|s, a) - q\|_1 \leq \beta(s, a) \right\} \quad (4)$$

$$\mathcal{C}'(s, a) = \left\{ \lambda \in [0, 1] : |\widehat{R}(s, a) - \lambda| \leq \beta'(s, a) \right\} \quad (5)$$

Then, the first maximization in EVI is straightforwardly solved as

$$\max \{ \lambda \in \mathcal{C}'(s, a) \} = \widehat{R}(s, a) + \beta'(s, a)$$

The second maximization in EVI does not admit a closed-form solution in general, but can be solved efficiently using only $\mathcal{O}(S)$ operations. Algorithm 3 below presents a simple procedure to find

$$q^* \in \operatorname{argmax} \left\{ \sum_{x \in \mathcal{S}} q(x) u(x) : q \in \mathcal{C}(s, a) \right\} \quad (6)$$

for a generic set $\mathcal{C}(s, a)$.

Algorithm 3 Procedure for solving problem (6) [2]

initialization: Let the elements of $\mathcal{S} = \{s_1, s_2, \dots, s_S\}$ be indexed such that $u(s_1) \geq \dots \geq u(s_S)$.
Set $\begin{cases} q^*(s_1) = \min\{1, \widehat{P}(s_1|s, a) + \frac{1}{2}\beta(s, a)\} \\ q^*(s_j) = \widehat{P}(s_j|s, a), \quad j = 2, \dots, S \end{cases}$
Set $\ell = S$
while $\sum_{x \in \mathcal{S}} q^*(x) > 1$ **do**
 Set $q^*(s_\ell) = \max\{0, 1 - \sum_{x \neq s_\ell} q^*(x)\}$
 $\ell = \ell - 1$
end while

Some considerations for implementation. We provide two practical considerations that could prove useful when implementing UCRL2:

- Initialize $\widehat{P}(\cdot|s, a) = \frac{1}{S}$.
- Limit the number of iterations for EVI to some sufficiently large number (e.g., 1000). In general, this is not necessary for EVI, but in early rounds, as only few data may have been observed, the convergence criterion of EVI may not be satisfied in finite iteration.

4 UCRL2-L: A Variant of UCRL2 with Sharper Confidence Sets

The confidence sets in UCRL2 are built using Weissman's and Hoeffding's inequalities combined with union bounds. The confidence sets can be made sharper by resorting to using more advanced methods to handle the random number $N_t(s, a)$ of observations. In particular, Appendix A presents a *time-uniform* variant of Weissman's inequality in which the randomness in the number n of samples is handled by *the method of mixtures* ([4]). This leads us to devise a variant of UCRL2, called UCRL2-L [5], which uses the following confidence sets: For each $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$\mathcal{C}_{t, \delta}^{\text{UCRL2-L}}(s, a) = \left\{ P' \in \Delta(\mathcal{S}) : \|\widehat{P}_t(\cdot|s, a) - P'\|_1 \leq \beta_{N_t(s, a)}\left(\frac{\delta}{SA}\right) \right\}, \quad (7)$$

$$\mathcal{C}'_{t, \delta}^{\text{UCRL2-L}}(s, a) = \left\{ R' \in [0, 1] : |\widehat{R}_t(s, a) - R'| \leq \beta'_{N_t(s, a)}\left(\frac{\delta}{SA}\right) \right\}, \quad (8)$$

where for $n \in \mathbb{N}$,

$$\beta_n(\delta) := \sqrt{\frac{2}{n} \left(1 + \frac{1}{n}\right) \log\left(\sqrt{n+1} \frac{2^S - 2}{\delta}\right)}, \quad \beta'_n(\delta) := \sqrt{\frac{1}{2n} \left(1 + \frac{1}{n}\right) \log\left(\frac{\sqrt{n+1}}{\delta}\right)}.$$

We further define

$$\mathbf{C}_{t,\delta}^{\text{UCRL2-L}} = \cap_{s,a} \mathbf{C}_{t,\delta}^{\text{UCRL2-L}}(s, a), \quad \mathbf{C}_{t,\delta}'^{\text{UCRL2-L}} = \cap_{s,a} \mathbf{C}_{t,\delta}'^{\text{UCRL2-L}}(s, a). \quad (9)$$

To our best knowledge, UCRL2-L was first introduced in [5] as a baseline method. Its pseudo-code is similar to that of UCRL2, and the only difference is the width of the confidence sets. Nonetheless, in practice, UCRL2-L significantly outperforms UCRL2, and often by an order of magnitude.

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A Weissman’s Inequality

Weissman’s inequality controls the L_1 distance between a discrete distribution and its empirical estimation. More specifically, consider an alphabet $\mathcal{X}$ and let P be a probability distribution over $\mathcal{X}$. Let $\hat{P}_n$ denote the empirical estimate of P built using n i.i.d. samples from P . That is, given $X_1, \dots, X_n$ are drawn from P in an i.i.d. fashion, define $\hat{P}_n(x) := \frac{1}{n} \sum_{i=1}^n \mathbb{I}\{X_i = x\}$ for all $x \in \mathcal{X}$. Then, for all $\delta \in (0, 1)$, Weissman’s concentration inequality asserts that

$$\mathbb{P} \left(\|P - \hat{P}_n\|_1 \geq \sqrt{\frac{2}{n} \log \left(\frac{2^{|\mathcal{X}|} - 2}{\delta} \right)} \right) \leq \delta \quad (10)$$

Note that in (10) n is a deterministic number, and to construct a confidence set to be used in RL, we need to extend the above to make it valid for a random number n of samples. (Note that in RL, n is a random number depending on the history.) As discussed before, to this aim, one can use union bounds or more powerful tools such as *the peeling trick* [6] and *the method of mixtures* [4] to handle the randomness in n . In particular, using the above notations, Weissman’s inequality (10) combined with the method of mixtures asserts that

$$\mathbb{P} \left(\exists n \in \mathbb{N} : \|P - \hat{P}_n\|_1 \geq \sqrt{\frac{2}{n} \left(1 + \frac{1}{n} \right) \log \left(\sqrt{n+1} \frac{2^{|\mathcal{X}|} - 2}{\delta} \right)} \right) \leq \delta \quad (11)$$

The concentration inequality (11) is referred to as a *time-uniform* variant of Weissman’s inequality. We refer to [7] for other treatments.

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